Evidence Lab

Teaching Guide - Version 1.0

For Librarians, Faculty, and Instructors

Follow the evidence. Make the call.

Evidence Lab is an institution-neutral information-literacy investigation game for college students. This public guide explains how to teach with the game without revealing keyed answers, hidden evidence values, ideal paths, or case resolutions.

Play	https://yanlingliuinfolit.github.io/evidence-lab/play.html
Case catalog	https://yanlingliuinfolit.github.io/evidence-lab/cases.html
For educators	https://yanlingliuinfolit.github.io/evidence-lab/educators.html
Creator	Yanling Liu, Coordinator of Information Literacy and Research Services, Indiana University Kokomo

1. What Students Do

Each investigation follows a consistent cycle: Encounter -> Investigate -> Decide -> Explain -> Feedback -> Review. Students choose what evidence to inspect, decide when the evidence is sufficient, make a decision, identify the evidence that mattered most, reflect on confidence, and receive feedback. Students are not rewarded for opening every artifact.

2. Four Investigation Levels

Level 1 - Guided

Cases 1-4

Relatively clean evidence. Students practice verifying existence, comparing claims with sources, evaluating AI output, and calibrating uncertainty.

Core question: What should I check?

Level 2 - Networked

Cases 5-12

Students trace information through people, platforms, summaries, reposts, corrections, and access systems.

Core question: Where did this information actually come from?

Level 3 - Contextual

Cases 13-28

Students integrate provenance, visual context, authority, currency, health, money, careers, and institutional consequences.

Core question: What evidence matters for this decision in this context?

Level 4 - Synthesis

Cases 29-36

Students combine data interpretation, sampling, breaking-news uncertainty, source independence, translation, AI citation tracing, and bounded synthesis.

Core question: What is the strongest claim the evidence actually warrants?

3. Learning Outcomes

1. Verify provenance and existence of sources, images, quotations, accounts, opportunities, and claims.
2. Select evidence strategically rather than collecting information indiscriminately.
3. Evaluate claim-evidence fit and identify overstatement, causal shifts, and changes in scope.
4. Trace information networks and distinguish repetition from independent corroboration.
5. Evaluate authority contextually by considering expertise, purpose, relevance, platform, and currency.
6. Interpret visual, quantitative, and multimodal evidence.
7. Calibrate uncertainty and distinguish firm, qualified, and unresolved conclusions.
8. Update conclusions responsibly when new evidence becomes available.
9. Make consequential information decisions involving research, money, careers, health, and everyday life.
10. Communicate bounded conclusions without overstating or understating the evidence.

4. Investigator Points

Dimension	Points	Interpretation
Decision	4	Does the conclusion fit what the evidence warrants?
Evidence Quality	3	Did the student find evidence capable of resolving the question?
Reasoning	2	Does the evidence identified as most important support the

		conclusion?
Efficiency	1	Did the student stop after sufficient high-value evidence rather than checking indiscriminately?

Confidence is reflective and never changes the score. Evidence Lab scores are best interpreted formatively alongside investigation paths and discussion. The public game is not a validated standardized measure of information-literacy ability and should not be used as a secure high-stakes examination.

5. Session Modes

Practice Mode

Twelve balanced investigations are selected from Cases 1-35. Use Practice Mode for library instruction, course activities, workshops, independent practice, and low-stakes formative assessment.

Capstone Mode

Students complete eleven balanced investigations plus Case 36 as the finale. The capstone emphasizes synthesis and appropriately bounded communication rather than simply labeling information true or false.

6. Ready-to-Use Teaching Models

15-Minute Evidence Sprint

Choose 2-3 cases. Spend about 2 minutes orienting students, 8 minutes investigating, and 5 minutes comparing which evidence mattered and when there was enough.

50-Minute Session

5 minutes orientation; 25-30 minutes investigation; 10-15 minutes debrief; 5 minutes transfer to the current assignment or discipline.

Independent Practice

Assign a Practice session. Ask students to identify one case where their first impression changed, one strong artifact, and one low-value check.

Asynchronous Assignment

Assign a session before class and have students bring one Investigation Log example for discussion.

Capstone

Use Capstone Mode and ask students to restate the final claim in evidence-bounded language.

Faculty-Selected Mini-Set

Use the case catalog to select cases aligned with disciplinary goals. Avoid choosing only misleading cases; include supported/positive verification and, where useful, unresolved cases.

7. Course and Program Integration

Composition / First-Year Writing

AI citations, claim-source fit, summaries, research access, and synthesis.

First-Year Seminars

Evidence-based decision making across academic and everyday contexts.

Research Methods

Study design, health claims, data visualization, polling, causal language, and scope.

AI Literacy

Verification of AI-mediated claims rather than a simple pro/anti-AI frame.

Journalism / Media Studies

Source chains, visual context, breaking news, translation, source independence, and data.

Career / Professional Courses

Recruiters, internships, scholarships, job scams, and institutional messages.

8. Discussion and Debrief

- Which evidence most changed your decision?
- What looked useful but actually added little?
- When did you have enough evidence to stop?
- What would have to change for another answer to become defensible?
- Did the information look trustworthy before you investigated it?
- Did your confidence change after checking evidence?
- Where does this reasoning problem appear in your discipline or everyday life?

9. Using the Investigation Log

The Investigation Log is most useful as a reflection tool rather than a score report. Ask students to compare evidence-opening order, stopping decisions, confidence, and the evidence they identified as most important. Useful prompts include: What did you check first and why? What did you open that you did not need? When did you return for more evidence? Which case changed your mind?

10. Accessibility, Localization, and Simulation

Essential reasoning should not depend on color, image recognition, hearing, or knowledge of a language other than the instructional language. Visual cases should have meaningful text-equivalent evidence; video should have transcript/timeline support; charts should have data tables; multilingual cases should provide translations sufficient to investigate the case.

The default game requires zero institutional customization. A fictional campus is used where a campus identity is useful, and library systems are described by function. Optional localization should never require students to know a local product name in order to succeed.

Many scenarios are fictional or composite instructional artifacts. Educators should preserve evidence-status disclosures and should not present simulated policies, alerts, opportunities, or people as real.

11. Selecting Cases

Select cases based on the reasoning move you want students to practice, not merely the topic. For a balanced mini-set, mix levels, environments, and outcomes. Include at least one case where investigation confirms information and, where appropriate, one case where uncertainty remains unresolved.

12. Spoiler-Free 36-Case Catalog

This catalog uses only student-public metadata. It intentionally excludes keyed answers, hidden evidence values, ideal paths, and case resolutions.

Case 1: Just Put It in the Bibliography

Level: Guided | **Environment:** 💬 Group Project Chat

Student goal: CITE IT?

Setup: Your group is finishing the bibliography for tomorrow's presentation.

Evidence status: SIMULATION

Case 2: Real Article, Bigger Claim

Level: Guided | **Environment:** 💬 Group Project Chat

Student goal: USE IT?

Setup: This time the group has a real article—but Mateo's sentence sounds much bigger than the study.

Evidence status: HYBRID

Case 3: ChatGPT Is Always Wrong

Level: Guided | **Environment:** 📚 Study Session

Student goal: USE IT?

Setup: Nia is trying to figure out where to start a research assignment.

Evidence status: GENERIC LIBRARY WORKFLOW + SIMULATION

Case 4: But It Says 100%

Level: Guided | **Environment:** 🤖 AI Conversation

Student goal: BELIEVE IT?

Setup: Omar asks an AI an obscure factual question for a club project.

Evidence status: SIMULATION

Case 5: I Didn't Read It, But...

Level: Networked | **Environment:** 💬 Before Class

Student goal: USE IT?

Setup: Min-jun is trying to catch up before discussion starts.

Evidence status: HYBRID TEMPLATE

Case 6: GUYS DID YOU HEAR??

Level: Networked | **Environment:** 📱 Group Chat

Student goal: SHARE IT?

Setup: A campus rumor suddenly lands in the group chat.

Evidence status: SIMULATION

Case 7: Everybody Is Saying It

Level: Networked | **Environment:** 📡 Across Your Feeds

Student goal: TRACE IT

Setup: By the next day, the same claim seems to be everywhere.

Evidence status: SIMULATION

Case 8: The Rumor Was True

Level: Networked | **Environment:** 💬 Student Chat

Student goal: ACT ON IT?

Setup: A few days later, Fatima posts something else in the chat.

Evidence status: SIMULATION — FICTIONAL CAMPUS

Case 9: You Already Shared It

Level: Networked | **Environment:** 🔔 Social Update

Student goal: CORRECT IT?

Setup: New evidence changes something Diego shared earlier.

Evidence status: SIMULATION

Case 10: I Found the Perfect Article...But I Can't Read It

Level: Networked | **Environment:** 📚 Library Research

Student goal: WHAT NEXT?

Setup: Nia finds an article that looks perfect for her paper.

Evidence status: GENERIC LIBRARY WORKFLOW + SIMULATION

Case 11: The Expert Is Outside Their Expertise

Level: Networked | **Environment:** 📱 Social Post

Student goal: TRUST IT?

Setup: Omar sends the group a confident post from a famous physician.

Evidence status: SIMULATION

Case 12: DON'T STUDY AFTER 10 PM

Level: Networked | **Environment:** 📱 Study Group

Student goal: SHARE IT?

Setup: Sofia drops a viral study-tip screenshot into the group.

Evidence status: SIMULATION

Case 13: Wait...Is That Here?

Level: Contextual | **Environment:** 📸 Social Image

Student goal: SHARE IT?

Setup: Diego posts a supposed current celebrity sighting.

Evidence status: SIMULATION

Case 14: The Cropped Protest Photo

Level: Contextual | **Environment:** 📰 Image + News

Student goal: INTERPRET IT

Setup: A viral crop appears to show protesters threatening a driver.

Evidence status: HYBRID

Case 15: Is This AI-Generated?

Level: Contextual | **Environment:** 🖼️ Synthetic Media

Student goal: VERIFY IT

Setup: Friends disagree about an unusual image and focus on visual 'tells.'

Evidence status: SIMULATION

Case 16: The Video Starts 12 Seconds Late

Level: Contextual | **Environment:** 🎥 Viral Video

Student goal: INTERPRET IT

Setup: A 14-second campus clip begins with one student shouting.

Evidence status: HYBRID

Case 17: SCIENCE PROVES MUSIC HELPS YOU STUDY

Level: Contextual | **Environment:** 📱 Social + Research

Student goal: SHARE IT?

Setup: A viral post says research has settled whether music improves studying.

Evidence status: HYBRID

Case 18: My Favorite Creator Says This Supplement Works

Level: Contextual | **Environment:** 📱 Influencer + Health

Student goal: BELIEVE IT?

Setup: A wellness creator says a supplement boosts focus and cites a study.

Evidence status: HYBRID

Case 19: This Creator Actually Knows Their Stuff

Level: Contextual | **Environment:** 📱 Social Video

Student goal: USE IT?

Setup: Omar wants to reject a careful research explainer solely because it is on social media.

Evidence status: HYBRID TEMPLATE

Case 20: The Headline Says Coffee Is Good Again

Level: Contextual | **Environment:** 📰 News + Health

Student goal: INTERPRET IT

Setup: A headline turns an observational health association into advice.

Evidence status: HYBRID

Case 21: Congratulations! You Won a Scholarship

Level: Contextual | **Environment:** 📧 Scholarship Email

Student goal: PAY IT?

Setup: Priya receives an unexpected scholarship award notice.

Evidence status: SIMULATION

Case 22: Your Account Will Be Locked Today

Level: Contextual | **Environment:** 📧 University-Looking Email

Student goal: PAY NOW?

Setup: Nia receives a polished urgent billing message.

Evidence status: SIMULATION — FICTIONAL CAMPUS

Case 23: A Recruiter Just DM'd Me

Level: Contextual | **Environment:** 💼 Recruiter DM

Student goal: PURSUE IT?

Setup: Mateo receives an unsolicited recruiter message.

Evidence status: SIMULATION

Case 24: $35 an Hour. Remote. No Experience.

Level: Contextual | **Environment:** 📱 Job Text

Student goal: TAKE IT?

Setup: Wei receives a high-paying remote job text.

Evidence status: SIMULATION

Case 25: The Internship Is Real...But Is This Listing?

Level: Contextual | **Environment:** 💼 Job Board

Student goal: APPLY HERE?

Setup: Sofia finds a real internship title on a third-party board.

Evidence status: SIMULATION

Case 26: Professor Moved the Exam Online

Level: Contextual | **Environment:** 💬 Course Chat

Student goal: WHAT NOW?

Setup: Diego says tomorrow's exam moved online, but the evidence initially conflicts.

Evidence status: SIMULATION — FICTIONAL COURSE

Case 27: CAMPUS CLOSED TOMORROW

Level: Contextual | **Environment:** 📱 Screenshot + Alert

Student goal: BELIEVE IT?

Setup: A screenshot says campus is closed tomorrow.

Evidence status: SIMULATION — FICTIONAL CAMPUS

Case 28: Official university Page Says It's Free

Level: Contextual | **Environment:** 🔎 Web Search

Student goal: CURRENT?

Setup: A student finds an official university page describing a service as free.

Evidence status: SIMULATION — FICTIONAL CAMPUS

Case 29: This Chart Looks Terrifying

Level: Synthesis | **Environment:** 📊 Social Chart

Student goal: INTERPRET IT

Setup: A post says campus parking complaints 'SKYROCKETED.'

Evidence status: SIMULATION

Case 30: 90% of Students Agree!

Level: Synthesis | **Environment:** 📊 Poll + News

Student goal: TRUST THE POLL?

Setup: A poll result looks suspiciously neat, and Wei dismisses it because the sample is 'only' about a thousand people.

Evidence status: SIMULATION

Case 31: BREAKING: Nobody Knows Yet

Level: Synthesis | **Environment:** 📰 Breaking News

Student goal: WHAT CAN WE SAY?

Setup: Multiple accounts explain a loud explosion differently while officials are still responding.

Evidence status: SIMULATION

Case 32: Five News Sites Can't All Be Wrong

Level: Synthesis | **Environment:** 📰 News Ecosystem

Student goal: CORROBORATED?

Setup: Five sites report the same claim.

Evidence status: SIMULATION

Case 33: Mom Sent This in the Family Chat

Level: Synthesis | **Environment:** 👨‍👩‍👧 Multilingual Family Chat

Student goal: VERIFY IT?

Setup: Wei's family chat forwards a message about a new student fee and asks Wei to check it.

Evidence status: SIMULATION

Case 34: The Screenshot Is Real—The Translation Isn't

Level: Synthesis | **Environment:** 🌐 International News

Student goal: TRANSLATE IT?

Setup: A real foreign-language news screenshot circulates with a dramatic English caption.

Evidence status: SIMULATION

Case 35: AI Search Quoted the Article

Level: Synthesis | **Environment:** 🤖 AI Search

Student goal: CITE THE CLAIM?

Setup: An AI search result gives a confident statistic and a real citation.

Evidence status: HYBRID TEMPLATE

Case 36: AI Tutor Beat the Professor!

Level: Synthesis | **Environment:** 🤖 AI + Research + Social

Student goal: EXPLAIN IT

Setup: A viral post turns a study into a sweeping claim about higher education.

Evidence status: HYBRID TEMPLATE

13. Downloadable Lesson Plans

- 15-Minute Evidence Sprint
- 50-Minute Evidence Lab Session
- Independent Practice Assignment
- Capstone Assignment
- Faculty Mini-Set Guide
- Asynchronous Assignment

Public lesson plans are available from the Evidence Lab For Educators page:
https://yanlingliuinfolit.github.io/evidence-lab/educators.html

14. Confidential Instructor Key

A separate Evidence Lab Instructor Key v1.0 contains case spoilers, keyed interpretations, hidden evidence values, ideal investigation paths, and case-specific debrief guidance. It is intentionally not included in the public repository, GitHub Pages site, LibGuide, or Zenodo release. Educators who receive the key should store it in access-controlled institutional storage.

15. Reuse, Attribution, and Citation

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Creator: Yanling Liu, Coordinator of Information Literacy and Research Services, Indiana University Kokomo.

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